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Project Synopsis On

"INDIAN BANKING PHISHING UPI FRAUD"

Submitted in partial fulfilment of the requirement for the project work for the degree of

BACHELOR OF COMPUTER APPLICATION (BCA)

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BANGALORE UNIVERSITY

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CERTIFICATE

This is to certify that the project synopsis entitled **"INDIAN BANKING PHISHING UPI FRAUD"** is submitted by Ms. Chinmayi H S (USN: U03BX24S0019) in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025-2026.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.

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Place: Bengaluru

Date: 26-10-2026

CASE STUDY

Social-Engineering Phishing Attack on Indian Banking Customers

An Analysis of the CERT-In Documented Campaign of August 2021

and Its Relevance to UPI and Digital-Payment Fraud

Academic Report

Prepared for educational and research purposes

Primary Source: Indian Computer Emergency Response Team (CERT-In)

Advisory dated 10 August 2021

Abstract

India's rapid adoption of internet banking, mobile banking and Unified Payments Interface (UPI) services has made digital financial transactions convenient but has also increased exposure to social-engineering attacks. This case study examines a real-world phishing campaign documented by the Indian Computer Emergency Response Team (CERT-In) in August 2021. In the incident, Indian banking customers were targeted through fraudulent SMS messages that created urgency by claiming that their bank accounts required KYC verification. Victims were directed to counterfeit banking websites designed to collect internet-banking credentials, mobile numbers and one-time passwords (OTPs). Attackers could then use the captured authentication information to gain access to victims' accounts and conduct fraudulent transactions.

The incident demonstrates that financial cybercrime does not always require direct compromise of a bank's core infrastructure. Instead, criminals can manipulate customers into voluntarily providing legitimate authentication information. The case is particularly relevant to India's UPI environment because similar social-engineering techniques are used in digital-payment fraud. The Reserve Bank of India (RBI) and the National Payments Corporation of India (NPCI) have repeatedly warned users against sharing OTPs, PINs and other sensitive information and against clicking suspicious links.

This study analyses the attack method, vulnerabilities, root causes, impact, detection and response mechanisms, mitigation strategies and lessons learned. It concludes that effective protection requires cooperation among customers, banks, payment providers, regulators and law-enforcement agencies.

1. Introduction

Digital banking has transformed the way people in India access financial services. Customers can transfer money, pay bills, manage accounts and make purchases without visiting a bank branch. UPI has further simplified digital payments by allowing users to make account-to-account payments through mobile applications.

However, increased digital dependence has also created opportunities for cybercriminals. Financial fraudsters increasingly use social engineering to manipulate users rather than attempting to break sophisticated banking infrastructure directly. Common techniques include phishing, smishing, vishing, fake KYC requests, malicious applications, fraudulent websites and deceptive payment requests.

The Reserve Bank of India (RBI) has specifically warned that digital-payment fraud can involve users being tricked into disclosing PINs, OTPs and passwords, opening links received through messages and emails, or downloading malicious applications. RBI has therefore

directed payment-system operators and participants to strengthen customer awareness through multiple channels and languages.

The National Payments Corporation of India (NPCI) similarly warns that fraudsters use social-engineering techniques based on fear, greed and ignorance. NPCI lists fake links, QR-code scams, malicious applications and threatening SMS or social-media messages among fraud patterns against which users should remain vigilant.

This case study focuses on a CERT-In-documented phishing campaign from August 2021. The campaign is particularly useful for academic analysis because CERT-In documented the attack sequence in detail, including the fraudulent SMS, counterfeit banking website, credential collection, OTP capture and subsequent fraudulent transaction stage.

The objective of this study is to understand how such an attack works, why users become vulnerable, what consequences can result, and what technical and organisational controls can reduce the likelihood and impact of similar attacks.

2. Incident Description

On 10 August 2021, CERT-In published an advisory concerning phishing websites hosted using the ngrok platform and targeting Indian banking customers. According to the advisory, malicious actors created websites that impersonated internet-banking portals of Indian banks. These websites were designed to collect sensitive information including internet-banking credentials, mobile numbers and OTPs.

The attack began with an SMS sent to a potential victim. The message used a KYC-related theme and created a sense of urgency by suggesting that the customer's bank account would be suspended if verification was not completed. The victim was directed to click a link contained in the message. Instead of reaching the legitimate banking website, the link opened a counterfeit banking page designed to resemble an internet-banking portal and requested login credentials.

Once the victim entered the credentials, the attacker could initiate an authentication process. The victim's legitimate phone then received an OTP. The victim, believing that the request was part of the legitimate banking login process, entered the OTP into the phishing page. The attacker captured the OTP and could then use the authentication information to access the victim's account and perform fraudulent transactions.

CERT-In's documented sequence therefore shows a complete social-engineering attack chain:

Fraudulent SMS → Fake banking website → Credential theft → OTP capture → Account access → Fraudulent transaction

An important point for this case study is that CERT-In documented the attack technique and potential fraudulent transaction process. The source should not be interpreted as evidence of a

particular rupee loss suffered by a named bank or individual. This distinction is important for maintaining factual accuracy in an academic report.

Table 1: Summary of Attack Stages

Stage	Action	Outcome
1. Smishing	Fraudulent KYC SMS with urgency (account suspension threat) and malicious URL	Victim clicks the link
2. Phishing	Counterfeit banking website hosted via ngrok platform	Victim believes the page is legitimate
3. Credential Theft	Collection of internet-banking username and password	Attacker obtains primary credentials
4. OTP Capture	Victim enters genuine OTP into the fake site	Second factor compromised
5. Account Access	Attacker logs into legitimate banking service	Unauthorised access achieved
6. Fraud	Fraudulent fund transfers or other transactions	Potential financial loss to victim

Source: Synthesised from CERT-In advisory (10 August 2021).

3. Attack Type / Vulnerability

3.1 Smishing

The first stage of the attack was smishing, which is phishing conducted through SMS. The fraudulent message attempted to imitate a legitimate banking communication. KYC is a particularly effective theme because customers know that banks perform KYC-related activities. The attacker therefore used a legitimate banking concept in a fraudulent context.

3.2 Phishing

The victim was redirected to a fake banking website. The website's purpose was to make the user believe that they were interacting with their bank. The attack demonstrates why customers should not rely solely on the visual appearance of a website. A page can look similar to a genuine banking portal while being controlled by an attacker.

3.3 Credential Theft

The phishing page collected internet-banking credentials. Once credentials are disclosed, the attacker can attempt to use them against the legitimate banking service.

3.4 OTP Theft

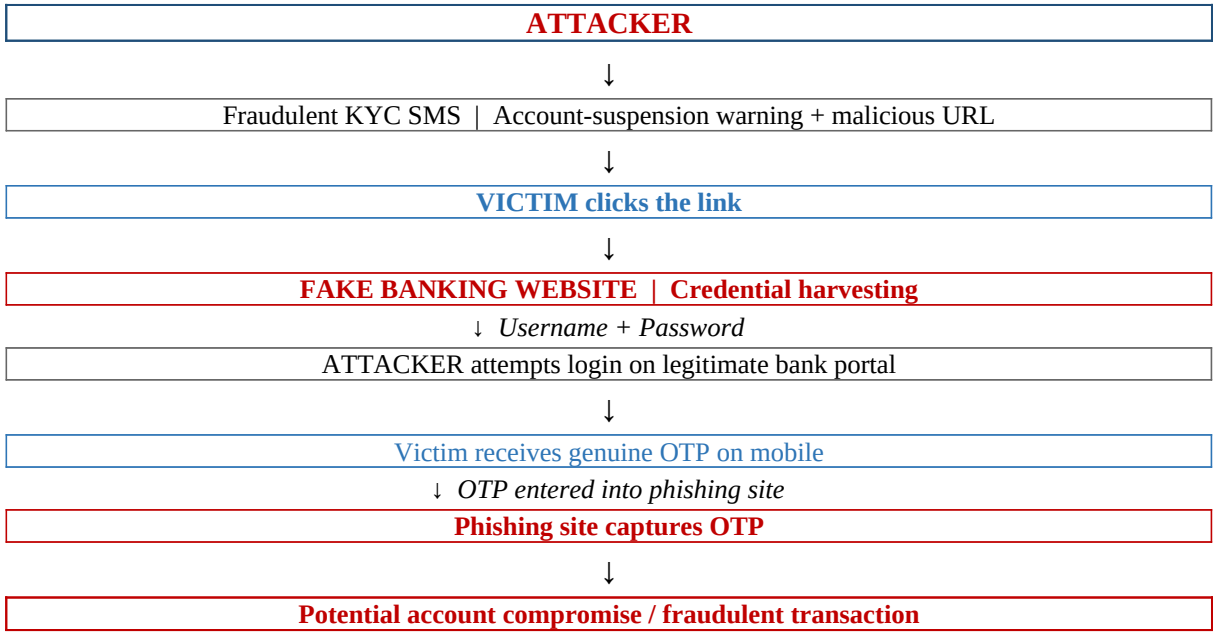
The second authentication factor provided an additional opportunity for the attacker. The victim received a genuine OTP but entered it into a fraudulent website. This demonstrates an important security principle: a strong authentication mechanism can still be defeated if the user is manipulated into providing the authentication factor to an attacker.

3.5 Infrastructure Abuse

CERT-In reported that the attackers abused the ngrok platform to host or expose phishing websites. The use of such services can make fraudulent infrastructure easier to deploy and can complicate detection and takedown.

4. Attack Flow Diagram

Figure 1: Phishing Attack Flow



Source: Original diagram created for this case study based on the attack sequence documented by CERT-In.

5. Root Cause Analysis

The incident did not result from one isolated technical weakness. It resulted from the interaction of technical and human vulnerabilities.

5.1 Human Behaviour

The attacker exploited the customer's trust in banking communications. The fraudulent message created fear by implying that the customer's account could be suspended. Under

pressure, a victim may act quickly without checking the destination URL or contacting the bank independently.

5.2 Lack of Independent Verification

A customer who receives an unexpected KYC message should independently access the bank's official application or website rather than clicking the supplied link. Failure to independently verify the communication increases the probability of successful phishing.

5.3 Authentication Information Disclosure

The victim provided a username, password and OTP to the fraudulent system. The authentication information itself may be technically valid, but its disclosure to an attacker defeats the purpose of the authentication mechanism.

5.4 Fraudulent Infrastructure

The attackers used fraudulent websites designed to resemble banking portals. CERT-In specifically reported the use of ngrok-hosted phishing infrastructure in the incident.

5.5 Awareness Gap

RBI has noted that digital-payment fraud continues despite awareness initiatives and has identified practices such as clicking links received through messages and disclosing sensitive payment information as recurring risks.

Therefore, the overall root cause can be represented as:

Social Engineering + Urgency + Credential Theft + OTP Disclosure + Fraudulent Infrastructure

Table 2: Root-Cause Categories

Category	Description
Human / Behavioural	Trust in banking SMS; response to urgency and fear of account suspension
Process / Verification	Failure to independently verify unexpected KYC requests via official channels
Authentication	Voluntary disclosure of username, password and OTP to a fraudulent site
Technical Infrastructure	Abuse of ngrok and similar services to host convincing phishing pages
Awareness / Education	Persistent gap between published warnings and day-to-day user behaviour

Source: Analytical synthesis based on CERT-In, RBI and NPCI materials.

6. Impact Analysis

6.1 Financial Impact

The most direct potential impact is unauthorised financial transactions. After obtaining the necessary authentication information, an attacker may attempt to access the victim's account and transfer funds. However, this case study does not assign a specific monetary loss to the CERT-In incident because the cited advisory does not provide a verified aggregate loss figure for the campaign.

6.2 Confidentiality Impact

The attack can expose sensitive information such as internet-banking credentials, mobile number, OTP and other information entered into the fraudulent portal. This information may also be reused in subsequent fraud attempts.

6.3 Account Security

Compromised credentials can potentially result in account takeover or unauthorised transactions.

6.4 Customer Trust

Repeated banking scams can reduce public confidence in digital banking. Customers may become uncertain about whether legitimate SMS messages, KYC requests and transaction notifications can be trusted.

6.5 Wider Digital-Payment Impact

The lessons from this incident apply to the wider digital-payment ecosystem. NPCI warns that fraudsters use social-engineering techniques involving fake links, QR codes, unknown applications and threatening messages. It also specifically warns that scanning a QR code and entering a UPI PIN is intended for making a payment, not receiving money. Therefore, although the CERT-In incident is primarily an internet-banking phishing case, its social-engineering characteristics are highly relevant to UPI fraud.

7. Detection & Response

Effective detection must occur at both the customer and financial-institution levels.

7.1 Customer-Level Detection

A user should consider the following warning signs:

- Unexpected KYC messages

- Threats of immediate account suspension
- Unknown or suspicious URLs
- Requests for banking passwords or OTPs
- Requests to install unknown applications
- Unexpected payment requests

A legitimate-looking message should not automatically be trusted.

7.2 Bank-Level Detection

Banks can use security analytics to identify suspicious activity, including:

- Login from a new device or unusual geographical location
- Multiple failed login attempts
- New beneficiary creation
- Unusual transaction amount or frequency
- Rapid movement of funds or behaviour inconsistent with the customer's normal pattern

RBI's digital-payment security directions include mechanisms for monitoring parameters such as failed transactions, transaction velocity, geolocation, IP information and behavioural indicators.

7.3 Immediate Response

If a customer discovers an unauthorised transaction, the first action should be to contact the bank immediately. The Government of India's National Cyber Crime Reporting Portal identifies 1930 as the immediate helpline for reporting cyber financial fraud. Victims should preserve SMS messages, screenshots, transaction IDs, bank statements, suspicious URLs, phone numbers, email messages and other relevant evidence. The Cyber Crime Reporting Portal also provides a facility for reporting suspicious website URLs and other cybercrime indicators.

8. Security Measures & Mitigation

8.1 Measures for Customers

1. Never share OTPs or UPI PINs with another person.
2. Avoid clicking banking links received through unexpected SMS messages.
3. Access banking services through the official application or manually entered official website.
4. Verify KYC requests independently.

5. Never install unknown banking or remote-access applications.
6. Enable transaction alerts and regularly check account statements.
7. Report suspicious transactions immediately.

8.2 Measures for Banks

Banks should implement real-time transaction monitoring, device-risk assessment, behavioural analytics, phishing-domain detection, suspicious beneficiary monitoring, risk-based authentication, automated transaction alerts, rapid blocking mechanisms and continuous customer awareness campaigns. RBI's customer-protection framework requires banks to maintain systems for electronic-transaction security, fraud detection and prevention, risk assessment and continuing customer awareness.

8.3 Measures for Payment-System Operators

Payment-system operators should monitor unusual transaction patterns and maintain anti-phishing and anti-rogue-application capabilities. RBI directions require payment-system operators to employ anti-phishing/anti-rogue-app services and establish monitoring mechanisms using factors such as transaction velocity, geolocation, IP origin and behavioural indicators.

8.4 Regulatory Measures

Regulators and government agencies should continue multilingual awareness campaigns, fraud-reporting mechanisms, threat-information sharing, phishing-site takedown coordination, and cooperation between banks, telecom providers and law enforcement.

Table 3: Multi-Layer Mitigation Framework

Stakeholder	Key Controls
Customers	Never share OTP/PIN; avoid unexpected links; use official apps; independent KYC verification; enable alerts
Banks	Real-time monitoring, behavioural analytics, risk-based auth, phishing-domain detection, rapid blocking
Payment Operators	Anti-phishing/anti-rogue-app services; velocity, geolocation and behavioural monitoring
Regulators / Govt	Awareness campaigns, 1930 helpline, threat sharing, coordinated takedowns

Source: Derived from RBI, NPCI and CERT-In guidance.

9. Lessons Learned

The incident provides several important cybersecurity lessons.

1. **Humans remain a major attack surface.** Even sophisticated banking systems can be undermined when a customer is manipulated into disclosing legitimate credentials.
2. **OTP does not mean automatic security.** An OTP provides additional authentication, but users must understand what transaction or login action they are authorising.
3. **Urgency is a powerful phishing technique.** Threats involving account suspension, KYC failure or immediate penalties can cause victims to act without verification.
4. **Phishing attacks can imitate trusted organisations.** A website that looks like a bank does not necessarily belong to the bank.
5. **Fraud prevention requires multiple layers.** Customer education, authentication, transaction monitoring, fraud analytics and rapid reporting must work together.
6. **Reporting speed matters.** RBI advises customers to notify their bank as soon as an unauthorised transaction is detected because delays can increase potential losses.

10. Recommendations

The following recommendations are proposed based on the analysis of the incident.

10.1 Improve Customer Awareness

Banks should conduct frequent awareness campaigns using SMS, mobile applications, websites, branches and regional languages. RBI has explicitly encouraged targeted multilingual awareness campaigns concerning safe digital-payment practices.

10.2 Strengthen Phishing Detection

Banks should continuously monitor domains, websites and applications that impersonate their brands.

10.3 Improve Transaction Monitoring

Financial institutions should use behavioural analytics to identify unusual account activity.

10.4 Use Risk-Based Authentication

High-risk transactions should receive additional scrutiny based on device, location, transaction value, beneficiary history and behavioural patterns.

10.5 Make Authentication Messages Clearer

OTP messages should clearly identify what action is being authorised whenever technically possible. RBI's digital-payment security directions include requirements concerning transaction identification in alerts and OTP-related communication.

10.6 Strengthen Inter-Agency Cooperation

Banks, NPCI, CERT , telecom operators and law-enforcement agencies should exchange threat intelligence rapidly when phishing campaigns are detected.

10.7 Simplify Fraud Reporting

Customers should have clear and easily accessible reporting options inside banking applications and websites. The national system provides the 1930 helpline for immediate reporting of cyber financial fraud.

10.8 Promote Secure Digital-Payment Behaviour

Users should understand basic principles such as:

- OTPs and UPI PINs are confidential.
- A UPI PIN is not required to receive money.
- Unknown links and applications should not be trusted or installed.
- Banking credentials should never be entered into suspicious websites.

11. Conclusion

The CERT-In-documented 2021 phishing campaign demonstrates how cybercriminals can exploit human behaviour to attack digital banking customers. Instead of directly breaking into a bank's core infrastructure, the attackers used a fraudulent KYC narrative to persuade customers to visit fake banking websites and provide credentials and OTPs. The captured information could then be used to access accounts and conduct fraudulent transactions.

The incident is significant because it illustrates the limitations of relying solely on technical authentication. Passwords and OTPs can provide strong protection when used correctly, but social engineering can cause users to disclose those credentials directly to attackers.

The same principles remain relevant to UPI and other digital-payment systems. NPCI and RBI continue to warn customers about fake links, social engineering, fraudulent applications, QR-code scams and disclosure of sensitive payment information.

Preventing banking and UPI fraud therefore requires a layered approach. Customers must develop secure digital habits, banks must strengthen behavioural and transaction monitoring, payment providers must improve fraud detection, and government agencies must provide rapid reporting and coordinated response mechanisms.

Ultimately, cybersecurity in digital banking is not only a technical problem. It is a combined technology, human behaviour, awareness and incident-response problem. The most effective defence is achieved when all parts of the financial ecosystem work together.

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Disclaimer

This case study has been prepared for academic and educational purposes only. It is primarily based on the real-world phishing activity documented by the Indian Computer Emergency Response Team (CERT-In) in August 2021. The report does not claim that every Indian bank or UPI user was affected, and it does not attribute a specific financial loss to a particular bank or individual unless supported by the cited evidence. The attack-flow diagram and analytical observations are an original academic synthesis of the documented incident. No real customer credentials, personal information or active malicious links are reproduced. Readers should use only official banking, government and law-enforcement channels when reporting suspected financial cybercrime.